

The effects of prenatal chronic noise exposure on rodent anxiety- and depression-like behaviours: protocol for a systematic review and meta-analysis

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Background

Noise pollution has emerged as a significant environmental stressor; its prevalence poses a health threat (Kjellberg 1990), particularly during the prenatal period (Barzegar et al. 2015). In the prenatal period, the brain is sensitive to stress-induced remodelling, with life long lasting consequences. This phenomenon is called early-life programming (Maccari et al. 2014), and it is associated with physical and psychiatric changes, including increased levels of anxiety and depressive behaviours (Maccari et al. 2014; Jafari et al. 2017). However, most human studies assessing the effect of noise in early-life programming are correlational, so experimental studies are needed establish causality (Salehpour et al. 2018).

Studies using rodent models in laboratory settings allow experimental control over noise exposure parameters and include behavioural and physiological assessments not possible in human studies (Robbins 2015). While some existing findings link prenatal acoustic stressors, such as ultrasonic tones and traffic noise, to anxiety-like behaviours (Barzegar et al. 2015), findings for depression-like behaviours show significant heterogeneity in magnitude. This variation is likely driven by differences in noise characteristics, rodent strains, offspring sex, and behavioural assays, and exposure timing (Badache et al. 2017; Abramova et al. 2021). This variation makes it difficult to draw general conclusions about the behavioural consequences of prenatal noise exposure. To address this issue, a systematic review and meta-analysis are required. This will provide a comprehensive, quantitative assessment of how chronic noise exposure influences anxiety- and depression-like behaviours in laboratory rodents and help identify the specific moderators driving the variation in outcomes.

Aim

Quantify the magnitude of the effect of maternal chronic noise exposure during gestation on offspring expression of anxiety- and depression-like behaviours in laboratory rodents and analyse key potential moderators of this effect.

Research questions

1. What is the overall effect of maternal exposure to chronic noise during gestation on anxiety and depression-like behaviours in offspring?
2. To what extent do biological factors (i.e., offspring sex or age class at measurement) moderate this effect?
3. To what extent do methodological factors—such as, exposure duration, noise type, behavioural assay type—are linked to the observed magnitudes of individual offspring responses to maternal chronic noise exposure?

Methods

Protocol and reporting

The reporting of search strategies adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses – Search extension (**PRISMA-S**) guidelines to support reproducibility of the search process (Rethlefsen et al. 2021). We will report the final report

following the PRISMA-EcoEvo (O'Dea et al. 2021), a set of reporting guidelines designed for meta-analyses in environmental and ecological research. The reporting of the methodology followed MeRIT to improve author contributions' granularity and accountability (Nakagawa et al. 2023a).

Research team

Our team includes researchers from various linguistic backgrounds (AL, SO, AM, ML, SN). We have broad experience in systematic evidence synthesis, meta-analysis, and are fluent in multiple languages (English, Spanish, Japanese, Polish, Russian).

Eligibility criteria

We developed eligibility criteria based on the Population, Exposure/Intervention, Comparator, Outcomes, and Study type (PECOS) framework to ensure clarity, consistency, and transparency throughout the screening process. **Table 1** summarizes the core PECOS elements used to guide study selection, and **Table 2** outlines the detailed inclusion and exclusion criteria.

Table 1. Population, Exposure/Intervention, Comparator, Outcome, Study type (PECOS)

framework elements defining the project scope.

PECOS	Evidence
Population	Offspring of laboratory-bred rodent dams that are non-genetically modified and free of pharmacological or surgical treatments, maintained under standard laboratory housing conditions following birth.
Exposure/ Intervention	Exposure to chronic noise administered to dams at any point during gestation (including exposure periods that may begin pre-gestation). Chronic noise refers to an intermittent or continuous acoustic stimulus, lacking musical structure (i.e., melody, rhythm, harmony), delivered over a prolonged period (i.e., ≥7 consecutive days). We will include any noise intensity (dB) that the study authors explicitly describe and use as a stressor.
Comparator	Laboratory-bred and non-genetically modified offspring of rodent dams maintained, under standard laboratory housing conditions and not exposed to noise during gestation (i.e., only exposed to ambient noise [including white noise], silence).
Outcome	Established measures of anxiety and depression measured in the offspring. Using one of these assays (or equivalent): Open Field Test (OFT), Elevated Plus maze (EPM), Light-dark box (LDB), Hole Board Test (HBT), Novelty Suppressed Feeding (NSF), Defensive Withdrawal Test (DWT), Tail Suspension Test (TST),

	Forced Swim Test (FST), Sucrose Preference Test (SPT).
Study type	Experimental primary studies, peer-reviewed or grey literature (theses).

Table 2. PECOS-structured inclusion and exclusion criteria

PECOS	Inclusion	Exclusion
Population	Offspring of laboratory-bred rodents (e.g., mice, rats, Guinea pigs) not genetically modified or pharmacologically or surgically treated animals), except sham controls.	Wild-caught rodents, rodents under the effect of behaviour modifying drugs (e.g., anxiolytics, antidepressants), except for sham pharmacological manipulations (e.g., saline). Rodents subjected to surgical procedures, except for sham interventions. Genetically modified rodents (e.g., transgenic, knockout/knock-in lines). Neither the offspring nor the dams should be subject to any other perinatal induced stress protocol beside chronic noise (e.g., restraining, crowding, maternal separation).
Exposure/ Intervention	Chronic noise exposure administered to dams exclusively during the prenatal (gestational) period (or before and during gestation). The exposure must	Exposure to music, defined as structured acoustic stimulation with organized pitch and/or rhythmic patterns (melody, beat, harmony), intentionally produced or recognized as music. Studies where chronic noise is combined with other

	meet the definition of chronic noise: an acoustic stimulus lacking musical structure, repeatedly delivered over an extended period (e.g., ≥ 7 consecutive days). Noise frequencies can be across infrasonic, audible, and ultrasonic ranges. Include studies using playback to simulate anthropogenic noise (e.g., urban, industrial noise), ultrasonic noise generators, synthetic computer-generated signals, loud white noise, or other acoustic stimuli recognized as a stressor by the authors.	unrelated stressors (e.g., electric shock, chemical exposure, lights) such that the specific effects of noise cannot be isolated. Acute noise (one-off or short exposures, e.g., tones to evaluate hearing thresholds or induce hearing loss). Playback of natural sounds (e.g., geotechnical, weather, other nature sounds) or other animals (e.g., predator calls, conspecific calls).
Comparator	Offspring from dams housed under laboratory standard housing conditions (no experimental noise, i.e.	Offspring of pregnant or lactating dams exposed to music or other structured or unstructured sounds considered stressors, or other experimental

Outcome	allowing conditions of ambient noise [including white noise], silence—if applied consistently across groups).	stressors, as a control condition.
	Include studies reporting at least one established measure of anxiety and depression in the offspring assessed postnatally, including: Anxiety: Open Field Test (OFT), Elevated Plus maze (EPM), Light–dark box (LDB), Hole Board Test (HBT), Novelty Suppressed Feeding (NSF), Defensive Withdrawal Test (DWT). Depression: Tail Suspension Test (TST), Forced Swim Test (FST), Sucrose Preference Test (SPT).	Outcomes limited solely to histological, biochemical, molecular, or genetic measures. Exclude studies that do not report any behavioural outcomes related to anxiety or depression, or studies that only tested the dams but not the offspring.
Study type	Experimental studies	Secondary literature (e.g., meta-analysis,

	published in peer-reviewed journals, as well as relevant grey literature (theses, dissertations).	systematic reviews), methodological articles, simulation studies. Case reports or studies without a control or baseline group. Abstracts and conference papers without detailed methods and results.
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Piloting

We piloted our literature search in multiple phases to ensure its feasibility, sensitivity, and transparency, following the relative recall approach (Lagisz et al. 2025). We compiled a benchmarking set of eight relevant primary experimental studies (**Table 3**) from Google Scholar, ResearchGate, and backward citation from a systematic review (Jafari et al. 2020). We piloted screening in Rayyan (Ouzzani et al. 2016) using 100 randomly selected records from Scopus to refine the PECOS eligibility criteria.

Table 3. Benchmarking set.

Benchmarking set	
1	Abramova O, Morozova A, Zubkov E, Ushakova V, Zorkina Y, Proshin AT, Storozheva Z, Gurina O, Chekhonin V (2024) Ultrasound-Induced Prenatal Stress: New Possibilities for Modeling Mental Disorders. <i>Dev Neurosci</i> 46:237–261. https://doi.org/10.1159/000534687
2	Abramova O, Ushakova V, Zorkina Y, Zubkov E, Storozheva Z, Morozova A, Chekhonin V (2021) The Behavior and Postnatal Development in Infant and Juvenile Rats After Ultrasound-Induced Chronic Prenatal Stress. <i>Front Physiol</i> 12:659366. https://doi.org/10.3389/fphys.2021.659366
3	Arjunan A, Sah DK, Rajan R (2023) Impacts of Maternal Noise Exposure on Risk of Stillbirth and Oxidative Stress-Induced Neurobehavioral Changes in Offspring. <i>Stresses</i> 3:529–540. https://doi.org/10.3390/stresses3030037
4	Barzegar M, Sajjadi FS, Talaei SA, Hamidi G, Salami M (2015) Prenatal exposure to noise stress: Anxiety, impaired spatial memory, and deteriorated hippocampal plasticity in postnatal life: PRENATAL SOUND STRESS AND COGNITION. <i>Hippocampus</i> 25:187–196. https://doi.org/10.1002/hipo.22363
5	Da Silva Oliveira P, Daniel AN, Nunes PT, Costa KA, Yehia HC, Ribeiro AM (2015) Intrauterine Exposure to Chronic 22 kHz Sound Affects Inhibitory Avoidance and Serotonergic Parameters in Forebrain Areas of Dams and Rat Offspring. <i>JBBS</i> 05:25–39. https://doi.org/10.4236/jbbs.2015.52003

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- 6 Jafari Z, Mehla J, Kolb BE, Mohajerani MH (2017) Prenatal noise stress impairs HPA axis and cognitive performance in mice. *Sci Rep* 7:10560.
<https://doi.org/10.1038/s41598-017-09799-6>
- 7 Pavlov D, Gorlova A, Haque A, Cavalcante C, Svirin E, Burova A, Grigorieva E, Sheveleva E, Malin D, Efimochkina S, Proshin A, Umriukhin A, Morozov S, Strekalova T (2023) Maternal Chronic Ultrasound Stress Provokes Immune Activation and Behavioral Deficits in the Offspring: A Mouse Model of Neurodevelopmental Pathology. *IJMS* 24:11712. <https://doi.org/10.3390/ijms241411712>
- 8 Uygur E, Arslan M (2010) Effects of chronic stress on cognitive functions and anxiety related behaviors in rats. *Acta Physiologica Hungarica* 97:297–306.
<https://doi.org/10.1556/APhysiol.97.2010.3.6>
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Search strategy

We developed the final search string using a combination of relevant keywords and Boolean operators (**Appendix 1**). We tested its sensitivity in Scopus against our benchmark set (Lagisz 2025). The string achieved 100% sensitivity, retrieving all 7/7 papers from the benchmark set indexed in Scopus (one paper was not indexed: doi 10.4236/jbbs.2015.52003). We adapted the Scopus search string to the other databases (Scopus, Web of Science, PubMed, OpenAlex, PsycINFO; **Table 4**) for peer-reviewed and grey literature, without restrictions on publication year or knowledge area.

We expanded our search to grey literature by using Bielefeld Academic Search Engine (BASE) and Google Scholar with simplified versions of our primary search string (**Table 4**). For Google Scholar, we will use Publish and Perish to get the first 100 most relevant records.

Although our search strings are in English, we will not apply language or region restrictions, so we expect to find records or publications in multiple languages. In such case, we will only consider records in the languages our team understand.

Table 4. Search strings for the full search strategy

Main search strings		
Scopus		
Date: Nov 7, 2025	Hits: 1403	Sensitivity: 100%
TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*" OR offspring OR "rat offspring" OR pup OR pups OR progeny) AND (anxiet* OR distress* OR despair* OR anhedonia OR "emotional state" OR depress* OR "anxiety-like" OR "depressive-like" OR "mood disorder*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla)))		
Web of Science Core Collection search string		
Date: Nov 7, 2025	Hits: 1840	Sensitivity: 100%
TS = ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*" OR offspring OR "rat offspring" OR pup OR pups OR progeny) AND (anxiet* OR distress* OR despair* OR anhedonia OR "emotional state" OR depress* OR "anxiety-like" OR "depressive-like" OR "mood disorder*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla)))		
PubMed		
Date: Nov 7, 2025	Hits: 1,055	
(("chronic noise"[tiab] OR "noise stress*" [tiab] OR "sound stress*" [tiab] OR "acoustic stress*" [tiab] OR "noise exposure"[tiab] OR "maternal noise exposure"[tiab] OR "gestational stress"[tiab] OR "prenatal stress"[tiab] OR "maternal stress"[tiab] OR "prenatal noise"[tiab] OR "ultrasound-induced stress"[tiab] OR ultraso* [tiab] OR "environmental noise"[tiab]		

OR "traffic noise"[tiab] OR "white noise"[tiab] OR "industrial noise"[tiab] OR "continuous noise"[tiab] OR "long-term noise"[tiab]) AND (prenatal[tiab] OR maternal[tiab] OR "during pregnancy"[tiab] OR "pregnant rat"[tiab] OR "pregnant rats"[tiab] OR "pregnant mice"[tiab] OR "pregnant dam"[tiab] OR "pregnant dams"[tiab] OR offspring[tiab] OR "rat offspring"[tiab] OR pup[tiab] OR pups[tiab] OR progeny[tiab]) AND (anxiet*[tiab] OR distress*[tiab] OR despair[tiab] OR anhedonia[tiab] OR "emotional state"[tiab] OR depress*[tiab] OR "anxiety-like"[tiab] OR "depressive-like"[tiab] OR "mood disorder*"[tiab] OR "open field"[tiab] OR "elevated plus maze"[tiab] OR "forced swim*"[tiab] OR "sucrose preference"[tiab] OR "light dark box"[tiab] OR "morris water maze"[tiab] OR "marble burying"[tiab]) AND (mouse[tiab] OR mice[tiab] OR rat[tiab] OR rats[tiab] OR rodent*[tiab] OR murine[tiab] OR "guinea pig"[tiab] OR chinchilla[tiab])) OR (((Rodentia [MeSH]) AND (noise[MeSH Terms])) AND (Pregnancy[MeSH Terms]))

PsycINFO

Date: November 13, 2025

Hits: 408

(chronic noise or noise stress or sound stress or acoustic stress or noise exposure or maternal noise exposure or gestational stress or prenatal stress or maternal stress or prenatal noise or ultrasound-induced stress or environmental noise or traffic noise or white noise or industrial noise or continuous noise or long-term noise).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word] AND (prenatal or maternal or during pregnancy or pregnant rat or pregnant mice or pregnant dam or offspring or rat offspring or pup or pups or progeny).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word] AND (anxiety or distress or despair or anhedonia or emotional state or depression or anxiety-like or depressive-like or mood disorder or open field or elevated plus maze or forced swim or sucrose preference or light dark box or marble burying).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word] AND (mouse or mice or rat or rats or rodent or murine or guinea pig or chinchilla).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh word]

OpenAlex

Date: November 7, 2025

Hits: 680

("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*" OR offspring OR "rat offspring" OR pup OR pups OR progeny) AND (anxiet* OR distress* OR despair* OR

anhedonia OR "emotional state" OR depress* OR "anxiety-like" OR "depressive-like" OR "mood disorder*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla)

Supplementary search strings

Google Scholar

Date: October 28, 2025 Hits: 489

("chronic noise" | "acoustic stress") ("prenatal stress" | "maternal stress" | offspring | pups) (anxiety | depression) (mice | rat | rodent)

BASE

Date: October 28, 2025 Hits: 2+2+5

noise rodent maternal doctype:18*

noise mice maternal doctype:18*

noise rats maternal doctype:18*

Screening

We will import the records from all search sources into R (R Core Team 2025) for initial de-duplication, then transfer them to Rayyan (Ouzzani et al. 2016) for a second de-duplication and a two-stage screening. The two-stage screening will start with the title+abstract stage, followed by a full text screening stage. To ensure methodological rigour and transparency, we will do an independent two-person screening (AL will be the primary screener, the rest of the team acting as second screeners) in both screening stages (title+abstract and full-text screening). We will resolve screening disagreements via discussion until agreement. During the full text screening stage, we will track and report the inclusion and exclusion decisions.

To support the consistent application of our eligibility criteria during screening, we developed a decision tree for the abstract and the full-text stages (**Figure 1**).

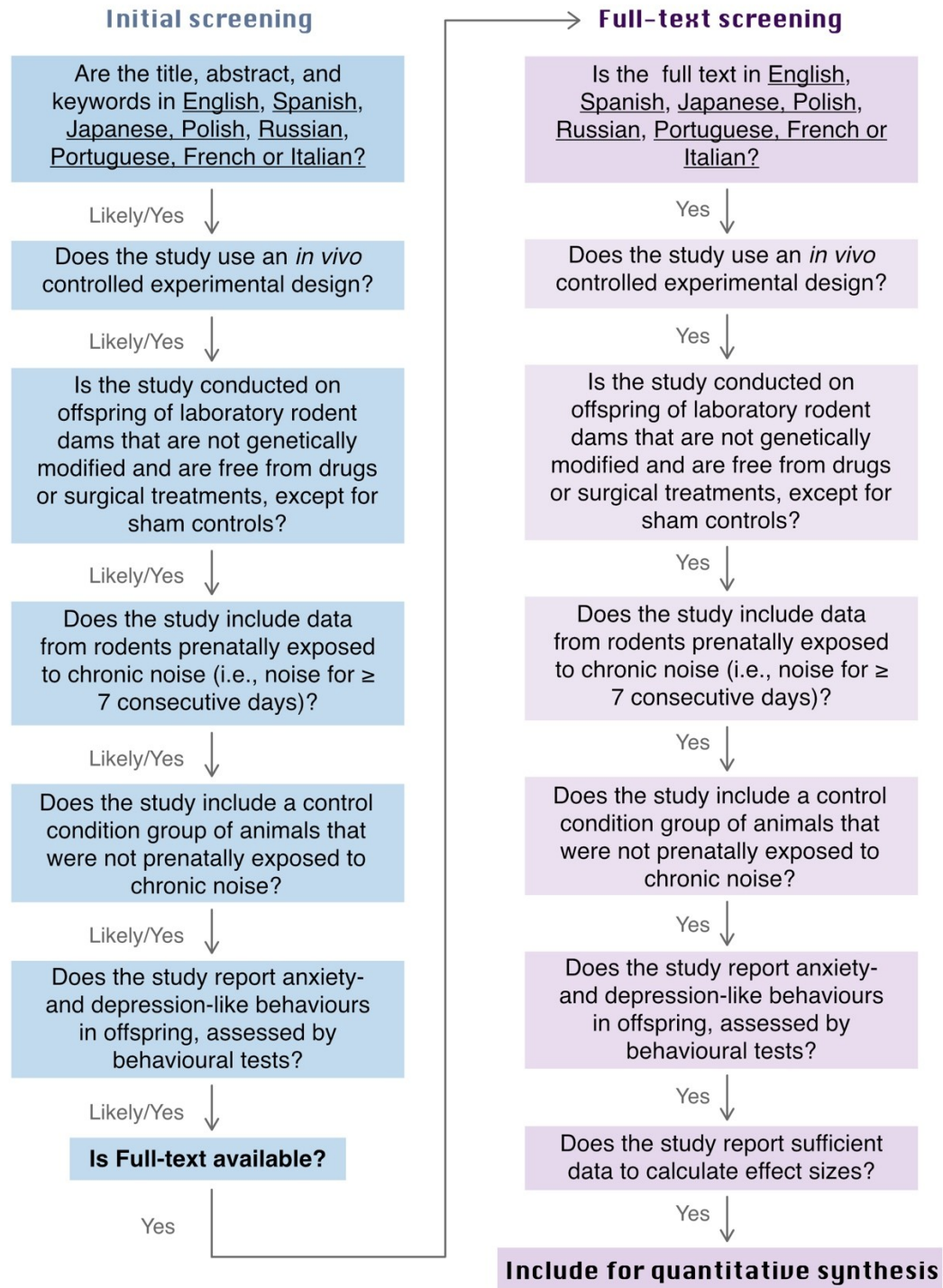


Figure 1. Screening decision tree flowchart at title, abstract and keywords and full text stages.

Data extraction

One team member will extract all the data (AL), and the supporting team members will cross-check 100% of the data extraction, dividing the data equally. We will extract the items listed in **Table 2**. If a study presents only graphical data, we will use metaDigitise (Pick et al. 2019) R package to extract the numerical data from the figures.

Two team members (ML and AL) independently extracted data from two of our benchmarking papers to pilot our data extraction variables. by (article DOIs: 10.1556/APhysiol.97.2010.3.6 and 10.1002/hipo.22363). After the extraction, we cross-checked the extraction and refined the variables accordingly.

Table 5. Extraction items

Items	Descriptions
Study information	
study_id	Unique identifier for the study. The format will be, in lower case, “first author_year_journal abbreviation.” E.g., smith_2023_jneurosc, with letters added for any non-unique IDs.
ex_cohort_id	Unique identifier for each independent experiment involving unique cohorts of animals within a study (e.g., one group exposed to 1 h of noise, another group exposed to 2 hours of noise), using a sequential numerical format (e.g., ex001, ex002). E.g., if a single cohort of animals is assessed using multiple different assays this is still considered a single experiment (ex001) with multiple outcome measures.
outcome_id	Unique identifier for each measurement assay or outcome reported for a specific cohort (e.g., out001, out002).
es_id	Effect size ID – A unique identifier for each control-experimental pair of measurements (exposure-control) used to calculate an effect size (e.g., es0001).
shared_control	Indicate if extracted ex_cohort_id shares a control group (e.i. c_a_id—see definition below) with another extracted effect size. Code “Yes” if they share a control group.
id_notes	Notes regarding the identifier for experimental cohorts of animals or other recorded IDs.
title	Title of the study.
doi	Short form DOI of the study (starting with “10.”).
published_in	The full name of the publisher. For journal articles, this will be the journal name; for theses, the university name.
study_notes	Comments regarding the study (e.g. atypical or complex study design requiring additional

explanations).

Population	
sp_latin	Latin binomial of the animal model used in the study (e.g., <i>Rattus norvegicus</i> , <i>Mus musculus domesticus</i>).
sp_common	The common name of the animal model used in the study (e.g. brown rat, house mouse)
strain	Strain/breed of the animal model used in the study (e.g., Wistar, Balb-6)
Dam	
dam_age	The dam's age at the start of the gestation, measured in weeks. Only numbers, if they provide a range of numbers, write the mid-point (e.g., 4-6 weeks = 5 weeks).
dam_age_notes	Any observations related to the dam's age, (e.g. mid-point recorded from the range 4-6 weeks).
gest_stage_ex	Gestation stage when the dams were exposed to chronic noise. For rats and mice, we will use the following categories: "early" stage (Day 0-7), "mid" gestation (Day 8-14), "late" gestation (Day 15 to term), or combination of these values. If not reported or unclear, code as "Not reported" or "Unclear", respectively. For other rodent species, we will categorize post-extraction according to their appropriate developmental stages.
gest_stage_ex_notes	Notes concerning the gestation stages at which the dams were exposed to chronic noise.
exposure_span	Total number of days the dam was exposed to the chronic noise. If it is not specified, code as "Not Specified".
exposure_type	Code the noise exposure as "Intermittent" if they played noise for some hours per days, or "Continuous" if they never turn the noise off during the total exposure duration.
exposure_session_duration	Only if the exposure was intermittent, is the number of daily hours of exposure (convert

minutes to hour fraction, e.g., 30 min is 0.5 h).

ex_session_duration _notes Notes related to the exposure timing (e.g., if the authors provided duration in minutes, write it here).

noise_type State the type of noise they used as the author states (synthetic noise, traffic playback, ultrasonic, other). For "other" add explanation in the noise_notes field.

loudness Loudness of the noise in decibels. Leave blank if not reported.

frequency Frequency of the speakers used, in hertz. Leave blank if not reported.

noise_notes Notes related to exposure conditions.

control_conditions Conditions under which the control animals were kept during the noise exposure phase: silence, ambient sound; white noise; other. For "other" add explanation in the control_notes field.

control_loudness Loudness of the control conditions in decibels, if provided.

control_notes Notes related to control conditions (e.g. no description provided).

Offspring

offspring_sex Sex of the offspring investigated: male, female, mixed, not reported. If not reported, the cell will be coded as "not reported".

offspring_age Age in days from the parturition of offspring at the start of a behavioural test. If they state, the age as days post conception (DPC), subtract 21 days. If they give a range of days, use the mid-point (e.g. 40-60 days, use 50 and add a note). If not reported or unclear, code as "not reported" or "unclear", respectively.

offspring_age_notes Notes concerning the age classes at which the animals went through the behavioural tests (e.g., if they provided a range of age, write it here).

experimental_procedures	Identifies if a sham control was used for a surgical or pharmacological intervention in the extracted groups of animals. Code as: "None" if no such intervention was performed. Code as "Sham", if a placebo procedure was used as the control (e.g., saline injection, mock surgery).
experimental_procedures_notes	Notes on the experimental procedures (e.g., type of sham procedure).
outcome_type	The primary behavioural construct to standardize whether an assay measures anxiety, depression, or both. For any assay listed in Appendix 2 , the outcome will be coded as either "Anxiety" or "Depression" as specified in that table (e.g., the Elevated Plus Maze is coded as "Anxiety"). If an assay is not in Appendix 2 , the classification explicitly provided by the study's authors will be used; in this case, the outcome can be coded as "Anxiety", "Depression", or "Both" based on their statement.
assay_type	Type of assay used to measure anxiety or depression-like behaviours as listed in Appendix 2 . Examples include Open Field Test, Tail Suspension Test, Elevated Plus Maze, Forced Swim Test, Sucrose Preference Test, Novelty Suppressed Feeding, Light-Dark Box.
assay_type_notes	Notes on the assay (e.g., variations from a typical protocol for a given assay type).
measurement_timing	The timing of the behavioural test. The value should be "Light Phase", "Dark Phase", or "Not Reported" if the information is not provided in the study.
measurement_variable	The specific behavioural phenomenon quantified in the assay. Examples include "Time in open arms," "Sucrose preference", "Total distance travelled", or "Immobility time".
data_type	The data type category of the measurement recorded. Examples include "Time," "Percentage," "Distance", "Count", or "Latency".
data_units	The specific unit for the measurement. For some data types like "Count", this may not be applicable (e.g., seconds "s", or centimetres "cm").
data_source	Location of extracted data in the publication. Indicate where the data (sample sizes,

averages, measures of dispersion) were reported (e.g., main text, table, figure, or supplementary material). If the source is a table or figure, specify its number (e.g., Table 1, Figure 1).

data_file	Name of the file where a screenshot of the figure with the extracted data is saved. If the data (sample sizes, averages, measures of dispersion) was shown in a figure, the screenshot should be saved in the format: First author_year_fig_number
co_a_id	Label given to the relevant control group (not exposed to noise in utero) by the authors (e.g., label used in a plot presenting the measured behavioural outcome).
ex_a_id	Label given to the relevant noise exposed group by the authors (e.g., label given in a plot presenting the measured behavioural outcome) .
comparison	Focal contrast label – A unique label should be created by concatenating the comparison ID (<i>co_id</i>) and the experiment ID (<i>ex_id</i>). For example: <i>control_chronicnoise</i> .
higher_better	Direction of outcome measurement – Indicates whether a higher mean value reflects lower levels of anxiety/depression. Options: “Yes” – Higher mean = less anxiety/depression; “No” – Higher mean = more anxiety/depression.
higher_better_notes	Notes on the direction of outcome measurement (e.g. shorter latency is better = less anxiety).
co_mean	Mean outcome measure for the offspring control group.
co_sd	Standard deviation for the offspring control group outcome measure.
co_n	Sample size for the offspring control group.
co_se	Standard error for the offspring control group outcome measure.
co_ci	Confidence intervals for the offspring control group outcome measure. If only CI are

	reported, they will be transformed to SE and noted in the stat_comment column.
ex_mean	Mean for the offspring experimental group outcome measure.
ex_sd	Standard deviation for the offspring experimental group outcome measure.
ex_n	Sample size for the offspring experimental group.
ex_se	Standard error for the offspring experimental group outcome measure
ex_ci	Confidence intervals for the offspring experimental group outcome measure. If only CI are reported, they will be transformed to SE and noted in the stat_comment column.
stat_comment	Notes regarding statistics extracted for the outcome measures.
general_comment	Any relevant overall notes for the extracted data.

Risk of Bias and Methodological Quality Assessment

We will evaluate the internal validity and methodological rigor of the included studies as quality-appraisal. For this, we adapted the **CRIME-Q tool** (Andersen et al. 2024), which provides a unified framework for assessing quality of reporting (QoR), methodological quality (MQ), and risk of bias (RoB) in animal research. CRIME-Q tool has been derived from the Collaborative Approach to Meta Analysis and Review of Animal Data from Experimental Studies (CAMARADES) checklist (Bahor et al. 2021), the ARRIVE 2.0 guidelines (Percie Du Sert et al. 2020), and SYRCLE’s risk of bias tool (Hooijmans et al. 2014), while also incorporating unique items, and providing flexibility to tailor the quality assessment criteria, items, and categories to the aims and focus of your synthesis, making it an ideal tool for our study.

Following the CRIME-Q tool guidance, we tailored their items to match our aims and focus.

Specifically, we removed items related to in-vitro procedures (items 2X and 2Y), as they do not apply to in vivo rodent behavioural tests. To streamline appraisal, we merged the four blinding-related items (7X, 7Z (1-3)) into a single item (7X) that specifies whether and how blinding was implemented. We also added seven QoR items related to transparency, as indicators of study quality. Our final assessment comprised 21 items, for which we developed predefined definitions (Appendix 3). Each item has explicit criteria that determine how we code responses into the answer categories (yes, no, partly, or unclear). We will conduct a two-person, independent appraisal of each study (focused on experiments and outcomes relevant to our aims), then discuss the differences, and if necessary, a third team member will help meet agreement. We will subject the independent assessment by two researchers to a Cohen's Kappa and report the disagreement rate for each item. Finally, we will present the results using a coloured grid representing the scores for each item and study. The CRIME-Q assessment will inform the interpretation of results and potential sensitivity analyses (Nakagawa et al. 2022).

Effect size calculation

We will use the log response ratio (lnRR), which is the natural logarithm of the ratio of the treatment mean to the control mean (Hedges et al. 1999; Lajeunesse 2015). We use the robust estimator of lnRR proposed by Nakagawa et al. (2023c).

Data synthesis and statistical analysis

We will use multi-level mixed-effects meta-analytic models to estimate the magnitude of effects of maternal chronic noise exposure on offspring anxiety- and depression-like behaviours in laboratory rodents. For each model, we will report confidence intervals and total heterogeneity. We will attempt to quantify the proportion of heterogeneity explained by moderator variables, conditional on the availability and distribution of relevant data. The potential moderators may include both biological (e.g., species/strain, sex, age) and methodological factors (e.g., timing and intensity of maternal noise exposure, noise type, measured outcome type, type of behavioural assay). We will test for publication bias using Egger's regression and funnel plot visualisation (Nakagawa et al. 2022). If publication bias is detected, we will explore its impact using sensitivity analyses. Where CRIME-Q quality assessments vary across studies, we may also include them as moderators in these analyses. We will summarize our results in text, tables, and visualise using forest or orchard plots along with bubble plots, as feasible (Nakagawa et al. 2023b).

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Appendix 1. Search string development

Scopus search strategy development history

Date	String id	Search string	Hits	Notes
14-Oct-25	pcne_1	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR "noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (gestation OR prenatal OR "pre-natal" OR maternal OR "pregnant dam*" OR "maternal exposure" OR "gestational exposure") AND (offspring OR pup* OR progeny OR juvenile* OR "neonatal" OR "postnatal day" OR "post-natal day") AND (depress* OR anxi* OR despair* OR anhedonia OR PTSD OR PPD OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "novelty suppressed feeding" OR "tail suspension" OR "defensive withdrawal" OR "hole board" OR "marble burying" OR "anxiety-like" OR "depression-like") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR gerbil OR hamster OR vole OR "guinea pig" OR chinchilla OR kunming))	24	
14-Oct-25	pcne_2	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR "noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (prenatal* OR gestation* OR maternal* OR "in utero" OR pregnan* OR dam OR dams OR	42	

		offspring OR pup OR pups) AND (depress* OR anxi* OR "anxiety-like" OR "depression-like" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "Novelty Suppressed Feeding" OR "tail suspension") AND (mice OR mouse OR rat OR rats OR rodent* OR gerbil OR hamster OR vole OR "guinea pig" OR kunming OR murine))		
14-Oct-25	pcne_3	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR "noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (prenatal* OR gestation* OR maternal OR "in utero" OR pregnan* OR dam OR dams OR offspring OR pup OR pups) AND (behav* OR anxi* OR depress* OR emotion* OR affective OR "anxiety-like" OR "depression-like" OR "mood disorder*" OR "emotional behav*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "Novelty Suppressed Feeding" OR "tail suspension") AND (mice OR mouse OR rat OR rats OR rodent* OR gerbil OR hamster OR vole OR "guinea pig" OR kunming OR murine))	82	
14-Oct-25	pcne_3_doi	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR "noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (prenatal* OR gestation* OR maternal OR "in utero" OR pregnan* OR dam OR dams OR offspring OR	4/7	Found: arjunan_2023, jafari_2017, barzegar_205, uygur_2010

		pup OR pups) AND (behav* OR anxi* OR depress* OR emotion* OR affective OR "anxiety-like" OR "depression-like" OR "mood disorder*" OR "emotional behav*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "Novelty Suppressed Feeding" OR "tail suspension") AND (mice OR mouse OR rat OR rats OR rodent* OR gerbil OR hamster OR vole OR "guinea pig" OR kunming OR murine)) AND DOI ("10.1038/s41598-017-09799-6" OR "10.3390/ijms241411712" OR "10.1159/000534687" OR "10.4236/jbbs.2015.52003" OR "10.1556/APhysiol.97.2010.3.6" OR "10.3390/stresses3030037" OR "10.3389/fphys.2021.659366" OR "10.1002/hipo.22363")		
14-Oct-25	pcne_4	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR "noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (prenatal* OR gestation* OR maternal OR "in utero" OR pregnan* OR dam OR dams OR offspring OR pup OR pups OR mice OR mouse OR rat OR rats OR rodent* OR gerbil OR hamster OR vole OR "guinea pig" OR kunming OR murine) AND (behav* OR anxi* OR depress* OR emotion* OR affective OR "anxiety-like" OR "depression-like" OR "mood disorder*" OR "emotional behav*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "Novelty Suppressed Feeding" OR "tail suspension"))	1,199	
14-Oct-25	pcne_4_doi	TITLE-ABS-KEY (("chronic noise" OR "noise exposure" OR	4/7	Found: arjunan_2023,

		"noise stress*" OR "acoustic stress*" OR "sound stress*" OR "environmental noise" OR "industrial noise" OR "urban noise" OR "traffic noise" OR "white noise" OR "ultrasonic noise" OR "audible noise" OR "infrasound" OR "acoustic overstimulation") AND (prenatal* OR gestation* OR maternal OR "in utero" OR pregnan* OR dam OR dams OR offspring OR pup OR pups OR mice OR mouse OR rat OR rats OR rodent* OR gerbil OR hamster OR vole OR "guinea pig" OR kunming OR murine) AND (behav* OR anxi* OR depress* OR emotion* OR affective OR "anxiety-like" OR "depression-like" OR "mood disorder*" OR "emotional behav*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "Novelty Suppressed Feeding" OR "tail suspension")) AND DOI ("10.1038/s41598- 017-09799-6" OR "10.3390/ijms241411712" OR "10.1159/000534687" OR "10.4236/jbbs.2015.52003" OR "10.1556/APhysiol.97.2010.3.6" OR "10.3390/stresses3030037" OR "10.3389/fphys.2021.659366" OR "10.1002/hipo.22363")	jafari_2017, barzegar_205, uygur_2010
14-Oct-25	pcne_5	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "gestational noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*") AND (anxiet* OR depress* OR "anxiety-like" OR "depressive-	1984

		like" OR "mood disorder*" OR neurobehav* OR "spatial memory" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "social behavior" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla OR offspring OR "rat offspring" OR pup OR pups OR progeny))	
14-Oct-25	pcne_5_doi	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "gestational noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*") AND (anxiet* OR depress* OR "anxiety-like" OR "depressive- like" OR "mood disorder*" OR neurobehav* OR "spatial memory" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "social behavior" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla OR offspring OR "rat offspring" OR pup OR pups OR progeny)) AND DOI ("10.1038/s41598- 017-09799-6" OR "10.3390/ijms241411712" OR "10.1159/000534687" OR "10.4236/jbbs.2015.52003" OR "10.1556/APhysiol.97.2010.3.6" OR "10.3390/stresses3030037" OR "10.3389/fphys.2021.659366"	7/7

		OR "10.1002/hipo.22363")		
14-Oct-25	pcne_5b	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "gestational noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*") AND (anxiet* OR depress* OR "anxiety-like" OR "depressive-like" OR "mood disorder*" OR neurobehav* OR "spatial memory" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "social behavior" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla OR offspring OR "rat offspring" OR pup OR pups OR progeny)))	1705	removed "social behavior", neurobehav*, spatial memory
14-Oct-25	pcne_5b_doi	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "gestational noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*") AND (anxiet* OR depress* OR "anxiety-like" OR "depressive-	7/7	

		like" OR "mood disorder*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla OR offspring OR "rat offspring" OR pup OR pups OR progeny)) AND DOI ("10.1038/s41598-017-09799-6" OR "10.3390/ijms241411712" OR "10.1159/000534687" OR "10.4236/jbbs.2015.52003" OR "10.1556/APhysiol.97.2010.3.6" OR "10.3390/stresses3030037" OR "10.3389/fphys.2021.659366" OR "10.1002/hipo.22363")		
22-Oct-25	pcne_5.1	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR "maternal noise exposure" OR "gestational stress" OR "prenatal stress" OR "maternal stress" OR "prenatal noise" OR "gestational noise" OR "ultrasound-induced stress" OR ultraso* OR "environmental noise" OR "traffic noise" OR "white noise" OR "industrial noise" OR "continuous noise" OR "long-term noise") AND (prenatal OR maternal OR "during pregnancy" OR "pregnant rat*" OR "pregnant mice" OR "pregnant dam*" OR offspring OR "rat offspring" OR pup OR pups OR progeny) AND (anxiet* OR depress* OR "anxiety-like" OR "depressive-like" OR "mood disorder*" OR "open field" OR "elevated plus maze" OR "forced swim*" OR "sucrose preference" OR "light dark box" OR "morris water maze" OR "marble burying") AND (mouse OR mice OR rat OR rats OR rodent* OR murine OR "guinea pig" OR chinchilla)))	1356	moved: offspring OR "rat offspring" OR pup OR pups OR progeny
22-Oct-25	pcne_5.1_doi	TITLE-ABS-KEY ((("chronic noise" OR "noise stress*" OR "sound stress*" OR "acoustic stress*" OR "noise exposure" OR	7/7	

"maternal noise exposure" OR "gestational stress" OR
"prenatal stress" OR "maternal stress" OR "prenatal noise" OR
"gestational noise" OR "ultrasound-induced stress" OR ultraso*
OR "environmental noise" OR "traffic noise" OR "white noise"
OR "industrial noise" OR "continuous noise" OR "long-term
noise") AND (prenatal OR maternal OR "during pregnancy" OR
"pregnant rat*" OR "pregnant mice" OR "pregnant dam*" OR
offspring OR "rat offspring" OR pup OR pups OR progeny) AND
(anxiet* OR depress* OR "anxiety-like" OR "depressive-like"
OR "mood disorder*" OR "open field" OR "elevated plus maze"
OR "forced swim*" OR "sucrose preference" OR "light dark
box" OR "morris water maze" OR "marble burying") AND
(mouse OR mice OR rat OR rats OR rodent* OR murine OR
"guinea pig" OR chinchilla)) AND DOI ("10.1038/s41598-017-
09799-6" OR "10.3390/ijms241411712" OR
"10.1159/000534687" OR "10.4236/jbbs.2015.52003" OR
"10.1556/APhysiol.97.2010.3.6" OR
"10.3390/stresses3030037" OR "10.3389/fphys.2021.659366"
OR "10.1002/hipo.22363")

Appendix 2. Common behavioural assays and outcomes

Common assays and outcomes to evaluate anxiety- and depression-like behaviours

Assay name	Primary behaviour assessed	Underlying principle/concept	Key measured parameters
Anxiety Assays			
Elevated Plus Maze (EPM)	Anxiety-like behaviour	Exploits innate aversion to open spaces and heights; approach-avoidance conflict	Time in open arms (%), entries into open arms (%), rearing, head dips, stretched-attend postures
Open Field Test (OFT)	Locomotor activity, exploration, anxiety-like behaviour	Exploits natural aversion to brightly lit open spaces and exploratory drive	Total distance travelled, line crossings, time/entries in centre area, rearing, stretch attend postures, sniffing
Light-Dark Box (LDB)	Anxiety-like behaviour	Exploits natural aversion to brightly lit areas (scototaxis) and exploratory drive; approach-avoidance conflict	Time in bright chamber (%), transitions, latency to enter light chamber
Novelty Suppressed Feeding (NSF)	Anxiety-related behaviour, antidepressant efficacy	Conflict between hunger drive and innate avoidance of novel, anxiogenic environment	Latency to eat in novel environment; home cage feeding, % weight lost
Hole Board Test (HBT)	Exploratory and anxiety-like behaviours	Exploits natural tendency to explore novel environments and holes in the floor	Number of head dips, duration of head dipping, latency to first head dip, rearing frequency, number of crossings
Defensive	Anxiety-like and defensive	Exploits the rodent's innate	Latency to exit dark chamber,

Withdrawal Test (DWT).	behaviours	preference for dark, enclosed space time spent in open field, (safety) versus aversion to brightly lit open areas (threat)	number and duration of entries in open area, stretch-attend postures, grooming or freezing episodes
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Depression Assays			
Sucrose Preference Test (SPT)	Anhedonia, motivation, depression-like emotional states	Measures loss of pleasure via reduced preference for palatable sweet solutions	Sucrose preference rate (sucrose consumption / total liquid intake) (%); total drinking times
Forced Swim Test (FST)	Behavioural despair, antidepressant efficacy	Measures immobility when escape is impossible, interpreted as “giving up”	Immobility time (s) (or mobility time)
Tail Suspension Test (TST)	Behavioural despair, antidepressant efficacy	Measures immobility when suspended by tail in inescapable situation	Immobility time (s) (or mobility time)

Appendix 3. CRIME-Q

Common assays and outcomes to evaluate anxiety- and depression-like behaviours

Items		Type	Questions	Criteria and Coding
0	study_id	ID	What is the unique identifier for the study?	The format will be, in lower case, "first author_year_journal abbreviation." E.g., smith_2023_jneurosc
0	title	ID	Title of the study	Copy the title of the study
CRIME-Q Items				
1X	peer_review	QoR	Did the paper undergo peer review prior to publication? Peer review might be useful for detecting errors or fraud. Yes/No	Code " Yes " if the journal is peer-reviewed. " No " if not (i.e., preprint)
1Xn	peer_review_notes	QoR	Any notes about the peer-review	Free text
3X	animals_reporting	QoR	Were the animals used in the experiment sufficiently described? Yes/Partly/No	Code " Yes " only if they report: dams' species/strain, vendor, dam age, housing (enrichment, number per cage), food type and availability, light/dark cycle, temperature/humidity. Offspring: sex, age, housing, enrichment, number per cage), light/dark cycle, temperature/humidity, food type and availability. If no information except species and strain, code as " No ". Otherwise, code as " Partly. "

3Xn	animals_reporting_note s	QoR	Any notes about the animals reporting	Free text
3Y	animals_methodology	MQ	Did the study use similar baseline characteristics for the animals in the control and exposed groups? And where the animals appropriate for the experiments? Yes/Partly/No	Our eligibility criteria will exclude non appropriate animals. However, only code "Yes" if they met the full criteria in animals reporting (3X). Studies with poor reporting will lack information to assess method quality. Add any relevant comment to animals_methodology_notes.
3Yn	animals_methodology_ notes	MQ	Any notes about the animal's- related methodology	Free text
3Z	animals_selection_bias (baseline characteristics) (SYRCLE Item 2)	RoB	Was the distribution of relevant baseline characteristics balanced between groups? Yes/No/Unclear	Code "Yes" if they provide evidence on how well they distributed the group (e.g., offspring male:female ratio, age, and weight equally distributed across groups. Code "No" if baseline characteristics do not match. Code "Unclear" if they don't provide relevant evidence.
3Zn	animals_selection_bias _notes	RoB	Any notes about selection bias	Free text
4Y	sample_size_calculatio n	QoR	Did the study include a calculation of sample size? Yes/Partly/No	Code "Yes" if they describe how they calculated the sample size- at what power, and if it was appropriate and well performed. Code "Partly" if sample size calculation is mentioned but not described in detail. Code "No" if power calculation is not

				mentioned.
4Yn	sample_size_calculatio n_notes	QoR	Any notes about sample size calculation	Free text
5X	in_vivo_reporting	QoR	Were the noise exposure and the behavioural tests sufficiently described: intensity and duration (whole experiment), assay details, etc., for the study to be fully transparent and reproducible? Yes/Partly/No	Code "Yes" if they describe in detail the exposure characteristics: if dams were exposed individually or in groups, the cage arrangement, number of dams per cage during the exposure, gestational stage, type of sound, intensity/frequency, duration, timing, and report the same information for the control dams. Additionally, they should describe offspring age at behavioural tests, which assays and the metrics they used. If any key details are missing, code as "Partly." If not details beyond the exposure and assay type, code as "No."
5Xn	in_vivo_reporting_note s	QoR	Any notes about the in-vivo reporting	Free text
5Y	in_vivo_methodology	MQ	Did the method seem feasible and technically well performed concerning the study's aim and outcome and in contrast to other known literature? Is it likely that the design influences the results— incomprehensive/insensible method? Yes/Partly/No	This question summarises items related to animals reporting (3X), animals' methodology (3Y), animals' selection bias (3Z), sample size calculation (4Y), and in vivo reporting (5X) (low transparency prevents the method quality assessment).

Code as **"Yes"** if the study fully meets the criteria(3X), methodology (3Y), selection bias (3Z), sample size calculation (4Y), and in vivo reporting (5X) (i.e. all **"Yes"**). Code as **"No"** if the study doesn't fully meet the criteria in items animals reporting (3X), animals' methodology (3Y), animals' selection bias (3Z), sample size calculation (4Y), and in vivo reporting (5X) (i.e. all **"No"**). Code as **"Partly"** if the study doesn't meet full criteria in any of the items animals reporting (3X), animals' methodology (3Y), animals' selection bias (3Z), sample size calculation (4Y), and in vivo reporting (5X).

5Yn	in_vivo_methodology_ notes	MQ	Any notes about the in-vivo methodology	Free text
5Z (1)	sequence_generation (SYRCLE Item 1)	RoB	Is there a description of sequence generation for allocation (the process by which experimental animals are assigned to experimental groups) And is it appropriate? Yes/No/Unclear	If the authors explicitly describe how dams were allocated to experimental groups and how offspring were allocated to outcome assays, code as "Yes" . If the authors do not mention allocation process and do not explicitly provide all sample sizes per group for dams and offspring, code as "No" . If only some sequence generation information is reported or

				are discrepancies code "Unclear").
5Z (1)n	sequence_generation_ notes	RoB	Any notes about the sequence generation	Free text
5Z (2)	random_housing (SYRCLE Item 4)	RoB	Were the animals randomly housed during the experiment? Yes/No/Unclear	Some experiments are influenced by the location of housing; hence random assignment of housing could negate these issues. The authors should describe animals' housing assignment outside periods of exposure. If the in vivo methodology (5Y) is incomplete, likely the random housing (5Z (2)) will be too. If they explicitly say they didn't house the animals randomly, code as "No." Otherwise, code as "Unclear."
5Z (2)n	random_housing_notes	RoB	Any notes about the random housing	Free text
5Z (3)	random_outcome_asse ssment (SYRCLE Item 6)	RoB	Were animals randomly selected for outcome assessments? For instance, if some endpoints (i.e., poor conditions, weight, etc.) were met and the investigators were not blinded, then the outcome cannot be assessed randomly. Not applicable to non-intervention studies. Yes/No/Unclear	Code "Yes" if the authors explicitly state that offspring were randomly selected for outcome assessment. Code "No" if the authors explicitly state that offspring were not randomly selected for outcome assessment (e.g. they were selected by body size or ocndition). Otherwise, code as "Unclear."
5Z (3)n	random_outcome_asse ssment_notes	RoB	Any notes about the random outcome assessment	Free text

Protocol v.4, November 14, 2025

6X	welfare_compliance	QoR	Did the study comply with any animal welfare regulations? Yes/Partly/No	Code as " Yes " if authors explicitly state they complied with the welfare regulations from an appropriate nod body in charge of ethics and animal welfare regulations
6Xn	welfare_compliance_n	QoR	Any notes about the welfare regulations compliance	Paste what the author said about regulations (e.g., "The Ethic Committee of X University approved the experimental protocol of the study.")
7X	blinding	QoR	Was the study blinded in any way, e.g. were the investigator or animal handlers blinded to the experimental group identity of the animals? Yes/No/Unclear	Blinding is a strategy for reducing the risk that researchers, animal care staff, and others involved may influence outcomes (subconsciously or otherwise). Code as " Yes " if the authors explicitly mention they did use blinding for any relevant procedures. Code as " No " if the authors explicitly mention they did not use blinding for all relevant procedures. If blinding is not mentioned, code as " Unclear. "
7Xn	blinding_notes	QoR	Any notes about the blinding	Free text
8X	methods_results_congruency	QoR	Did the study present all their findings based on the methods described? Is there congruency between the method and results sections? And is it transparent?	Presenting results in which methods are not described is not transparent and replicable and should be interpreted with caution. When key methodological details are missing or unclear (e.g., group

			Yes/Partly/No	allocation/selection, dam characteristics, offspring age, timing of behavioural tests, or noise-exposure parameters) the study cannot score "Yes." Without sufficient reporting, methods cannot be reliably matched to the measured outcomes and reported results and should be coded as "Partly" , unless all the results cannot be matched to the methods (e.g. when methods descriptions are missing) and are coded as "No."
8Xn	methods_results_congruency_notes	QoR	Any notes about methods and results congruency	Free text
8Z (1)	attrition_bias (incomplete outcome data) (SYRCLE Item 8)	RoB	Describe the completeness of outcome data including attrition and exclusions from the analysis and were incomplete outcome data adequately described? Were all animals included in the analysis and if not, was it described why they were not included?	If multiple Section 5 items are scored Partly or Unclear, it is unlikely the study will meet attrition_bias (8Z (1)). To code "Yes" , authors must explicitly report the starting number per group and the final analyzed sample size for each test/result/measurement and provide counts and reasons for any exclusions. Code as "Unclear" if sample sizes are missing for the outcome data. Code as "No" if sample sizes are not matching and

Yes/No/Unclear

				are not explained.
8Z (1)n	attrition_bias_notes	RoB	Any notes about the attrition	Free text
8Z (2)	reporting_bias_selective_outcome (SYRCLE Item 9)	RoB	Was the study protocol available (require a description of protocol location) and were all the study's prespecified primary and secondary outcomes reported in the manuscript? Yes/No/Unclear	If protocol is available and all outcomes match, code as "Yes." If there is no protocol or is not publicly available, code as "Unclear." If there is a protocol and does not match, code "No."
8Z (2)n	reporting_bias_selective_outcome_notes	RoB	Any notes about the reporting bias selective outcome	Free text
9X	limitations	QoR	Did the study contain a section of limitations, or did they comment on the limitations of the study in relationship to our focus outcomes in the rodents? Yes/Partly/No	If they have a specific study limitation section or discuss limitation of the outcomes of interest, code "Yes." If they briefly mention some study limitations, but not all code as "Partly." Code "No" if they don't mention any limitations.
9Xn	limitations_notes	QoR	Any notes about study limitations	Free text
10X	statement_conflict_of_interest	QoR	Did the study contain a statement of potential conflicts of interest? Yes/No	If there is a specific section stating Conflict of interest code as "Yes" (copy and paste statement in conflict-of_interest_notes). If there is not a section code as "No."
10Xn	statement_conflict_of_interest_notes	QoR	Copy the conflict-of-interest statement	Free text
10Z	publication_bias (SYRCLE Item 10)	RoB	Was the study free of inappropriate influence from funders or	If there is a specific section or text fragment stating such potential

			companies supplying drugs or equipment? Yes/No	influence *including commercial funding), code as "Yes" (copy the relevant text in publication_bias_funder_influence_notes). Code as "No" if there is no relevant information.
10Zn	publication_bias_notes	RoB	Copy relevant text	Free text
Extra items				
11X	Authors contributions	QoR	Is there an author contributor statement? Yes/No	Code "Yes" if there is a specific section with authors contribution, "No" otherwise
11X_n	authors_contributions_notes	QoR	If there is an author's contribution statement	Copy the statement
12X	analysis_software	QoR	Did the authors report the software they used for the analysis? Yes/No/NA	Code "Yes" if they reported the name of the software. "No" otherwise
12X_n	analysis_software_notes	QoR	If they reported the software for the analysis	Copy the name of the software
13X	data_availability	QoR	Are the raw data, coding and scripts publicly available? Yes/No/NA	If they used a software and not a script, code as "NA"
13X_n	data_availability_notes	QoR	If they shared their data, code and scripts	Copy from the text what the author said about the data and code availability